NUMERICAL METHODS WITH C++

Project Report

Interpolating an Implied Volatility Surface

Using Natural Cubic Splines

Nour SALHI

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Abstract

Implied volatility is one of the most important quantities in option pricing and risk management. However, market data is only available at discrete strike prices and maturities. Therefore, interpolation techniques are required to estimate implied volatility at points not directly quoted by the market.

The objective of this project is to estimate the implied volatility corresponding to a target maturity and strike price using a two-step natural cubic spline interpolation procedure. Option-chain data were collected from Yahoo Finance, cleaned, and organized into a structured dataset.

Interpolated Implied Volatility	
0.8329 (83.29%)	**Target Maturity:** 900 days
σIV (T* = 900d, K* = 108)	**Target Strike:** K* = 108
	Method: Natural Cubic Spline
	Observations: 49 valid data points

1. Introduction

The implied volatility surface is a three-dimensional representation showing how implied volatility varies with both strike price and maturity. Unlike the Black-Scholes model assumption of constant volatility, real financial markets exhibit volatility smiles and skews, making implied volatility dependent on option characteristics.

Mathematically, the surface is described as:

```
σ_IV = f(T, K)
```

where:

- T = time to maturity
- K = strike price
- σ_IV = implied volatility

The objective of this project is to estimate the implied volatility at:

- Target maturity: T* = 900 days
- Target strike: K* = 108

using natural cubic spline interpolation.

2. Data Collection

The data were collected from Yahoo Finance using call option contracts. The collection date used for all observations was 07 June 2024. The maturity was computed as:

```
Days to Maturity = Expiration Date - Collection Date
```

Implied volatilities reported by Yahoo Finance were converted from percentages to decimal form (e.g. 24.35% → 0.2435). After cleaning, 49 valid observations remained.

Variable	Description
Strike Price	Exercise price of the option
Expiration Date	Contract maturity date
Days to Maturity	Calculated maturity in calendar days
Implied Volatility	Market implied volatility (decimal)
Option Type	Call options only

3. Data Cleaning and Preparation

Several preprocessing steps were applied before interpolation.

3.1 Missing Values

Observations with missing IV values, invalid entries, or non-numeric observations were removed.

3.2 Conversion to Decimal Form

Yahoo Finance reports IV values in percentage form. These were converted to decimal format before processing.

3.3 Maturity Extraction

Expiration dates were extracted directly from contract names and converted into calendar days.

3.4 Sorting

The dataset was sorted according to (1) days to maturity and (2) strike price, to guarantee correct spline construction.

4. Theoretical Framework

4.1 Natural Cubic Splines

A cubic spline is a piecewise cubic polynomial connecting data points smoothly. For each interval $[x_i, x_{i+1}]$ the spline is:

```
S_i(x) = a_i + b_i(x-x_i) + c_i(x-x_i)^2 + d_i(x-x_i)^3
```

The spline satisfies three sets of conditions:

Continuity Conditions

1. Function continuity: $S_i(x_{i+1}) = S_{i+1}(x_{i+1})$
2. First derivative continuity: $S'_i(x_{i+1}) = S'_{i+1}(x_{i+1})$
3. Second derivative continuity: $S''_i(x_{i+1}) = S''_{i+1}(x_{i+1})$

Natural Boundary Conditions

Natural cubic splines impose zero second derivatives at the endpoints:

```
M_0 = 0    and    M_n = 0
```

These conditions prevent excessive oscillations at the boundaries.

5. Thomas Algorithm

To obtain the spline coefficients, a tridiagonal linear system must be solved. The system takes the form:

```
A_i * M_{i-1} + B_i * M_i + C_i * M_{i+1} = D_i
```

5.1 Forward Sweep

The coefficients are recursively transformed:

```
c'_i = C_i / (B_i - A_i * c'_{i-1})
d'_i = (D_i - A_i * d'_{i-1}) / (B_i - A_i * c'_{i-1})
```

5.2 Backward Substitution

```
M_n = d'_n
M_i = d'_i - c'_i * M_{i+1}
```

The algorithm runs in O(n) time — significantly faster than general matrix inversion at O(n^3).

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6. Two-Step Interpolation Procedure

Step 1 — Strike Direction Interpolation

For each maturity T_j, a natural cubic spline is constructed using strike prices as x-values. The observed points for maturity T_j are:

```
(K_{j,1}, σ_{j,1}),  (K_{j,2}, σ_{j,2}),  ...,  (K_{j,n}, σ_{j,n})
```

The spline is evaluated at K* = 108 to produce one intermediate volatility v_j per maturity.

Example — for maturity T = 896 days, the surrounding strikes are:

visualization_data

Maturity	Strike	ImpliedVolatility
896	105	0.8621
896	110	0.8241
896	115	0.8002
896	120	0.7601
896	125	0.748
896	130	0.7192
896	135	0.6994
924	85	0.9709
924	90	0.9216
924	95	0.8716
924	100	0.8626
924	105	0.8105
924	110	0.7965
924	115	0.7557
900	108	0.83292

Strike	Implied Volatility
105	0.862
110	0.824
115	0.800

Step 2 — Maturity Direction Interpolation

After Step 1, the reduced dataset becomes: {(T_1, v_1), (T_2, v_2), ..., (T_m, v_m)}. A second natural cubic spline is constructed using maturity as the x-variable. The final estimate is:

```
σ_IV(900, 108) = G(900)  =  0.8329
```

7. Results

The interpolation procedure generated the following result:

Parameter	Value
Target Strike	108
Target Maturity	900 days
Interpolated IV	0.8329
Percentage Form	83.29%

The resulting value lies between neighboring observations and is therefore consistent with market expectations.

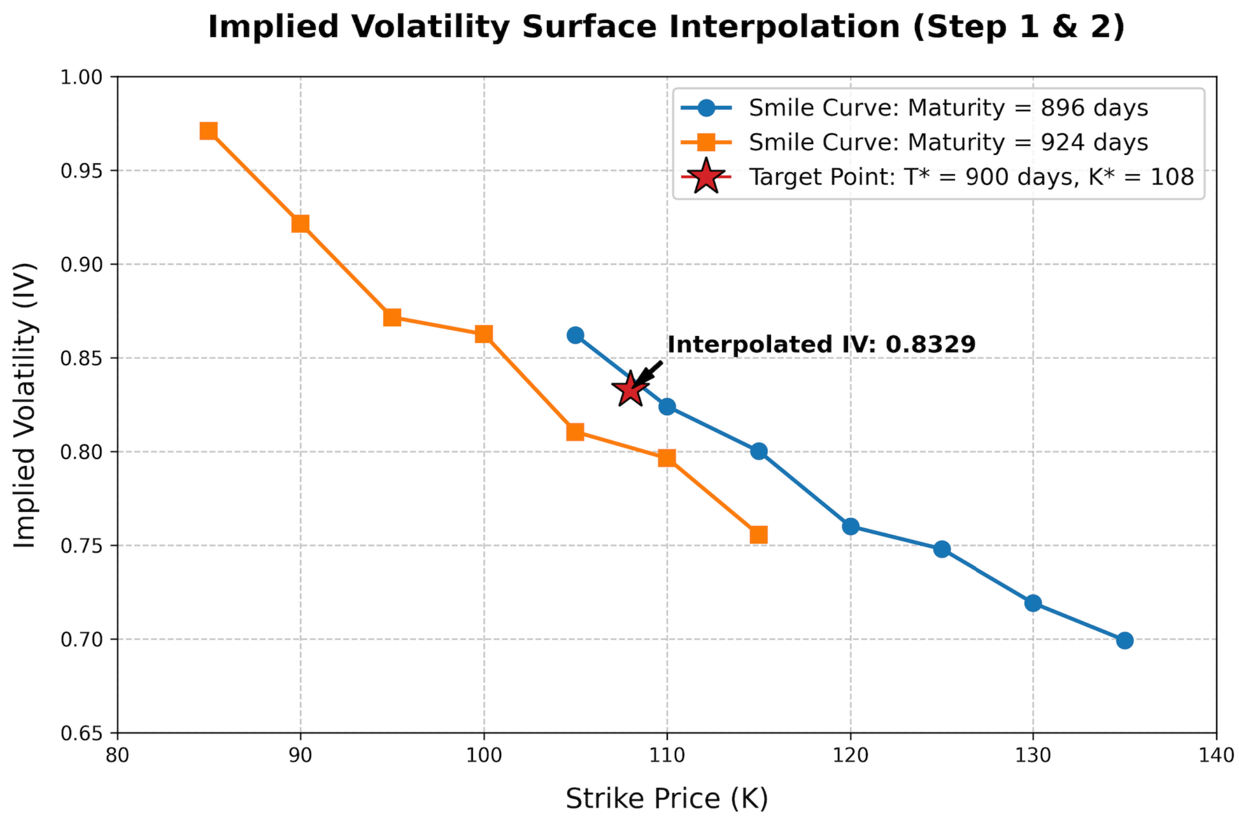
8. Financial Interpretation

The obtained volatility level of 83.29% indicates that market participants expect significant uncertainty regarding future price movements of the underlying asset.

The downward-sloping volatility smile suggests:

- Lower strikes are associated with higher implied volatilities.
- Investors are willing to pay a premium for downside protection.

This behavior is consistent with the volatility skew commonly observed in equity option markets.



9. Advantages of Cubic Splines

Advantage	Description
Smoothness	First and second derivatives remain continuous across all knots
Accuracy	Captures curvature that straight-line methods systematically miss
Stability	Natural boundary conditions suppress boundary oscillations
Financial Realism	Volatility surfaces are inherently curved — splines match that geometry

10. Limitations

Although spline interpolation is powerful, some limitations remain:

4. Results depend on data quality.
5. Illiquid options may produce noisy implied volatilities.
6. Market conditions can change rapidly.
7. Different interpolation techniques may generate slightly different estimates.
8. The method estimates volatility only within the observed data range.

11. Conclusion

This project successfully applied numerical methods to estimate an unobserved implied volatility using market option data. The methodology consisted of:

9. Collecting and cleaning Yahoo Finance option-chain data.
10. Constructing natural cubic splines across strike prices.
11. Applying a second spline across maturities.
12. Computing the implied volatility at the target point.

The final estimate obtained was:

```
σ_IV(900, 108) = 0.8329   (83.29%)
```

The result is consistent with surrounding market observations and demonstrates the effectiveness of natural cubic spline interpolation for constructing implied volatility surfaces. The project also highlights the practical application of numerical methods, tridiagonal systems, and the Thomas algorithm in quantitative finance.

References

13. Numerical Methods with C++ Lecture Notes.
14. Yahoo Finance Option Chain Data.
15. Hull, J. C. (2022). Options, Futures and Other Derivatives.
16. Wilmott, P. (2006). Paul Wilmott on Quantitative Finance.
17. Black, F. and Scholes, M. (1973). The Pricing of Options and Corporate Liabilities.